Variable Star Lab 1

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Abstract

In this lab, my lab partners and I observed a variable star and attempted to calculate its brightness as a Vega-system magnitude. The images we captured using the telescope needed to be corrected for variations in the telescope and sensor, and their data needed to be changed to units of counts of electrons per second. From this calibrated image, we calculated the counts per second of only the star being observed, which when compared to that of a calibration star of known brightness, allowed us to estimate the Vega-system brightness of our chosen variable star. This brightness came out to 13.696 ± 0.006 on October 4 and 12.180 ± 0.002 on October 6.

1 Introduction

For this lab, my group and I observed a variable star, meaning the star's brightness increases and decreases on a periodic basis. For the type of variable star that we chose (RR Lyrae), that period of brightness oscillations is linearly related to the star's luminosity. With that knowledge, our goal for this lab was to calculate the brightness of the variable star we chose as a Vega-system magnitude. If we did this on several nights over a longer time, we would be able to use these magnitudes to calculate the period and then use the period-luminosity relationship to calculate the star's luminosity, which could then be used to calculate its distance. However, this is beyond the scope of this experiment, which just aims to calculate the star's Vega-system magnitude for each of the two nights we observed it.

2 Observations

2.1 Instrument

The telescope used to take this data was the Nickel one-meter telescope at Lick Observatory and its Direct Imaging Camera, which uses a ccd sensor to capture images. The exposures were be taken using its Bessel V-filter to view wavelengths of light between about 480 and 600 mm

2.2 Targets

The science target my group chose for this lab was V0509 Peg, a star with right ascension 22:13:36.57 and declination +26 46 45.86. Its approximate Vega-system magnitude is 12.86.

The calibration field my group chose for this lab was SA41, with right ascension 21:53:51.46 and declination +45 25 45.5. We specifically used the star SA41-654 for calibration, which has a known Vega-system magnitude of 10.038.

2.3 Summary of Observations

The science star was exposed five times for 50 seconds on Oct. 4 and another five times for 55 seconds on Oct. 6. The calibration field was exposed five times for 15 seconds on Oct. 4 and another five times for 85 seconds on Oct. 6.

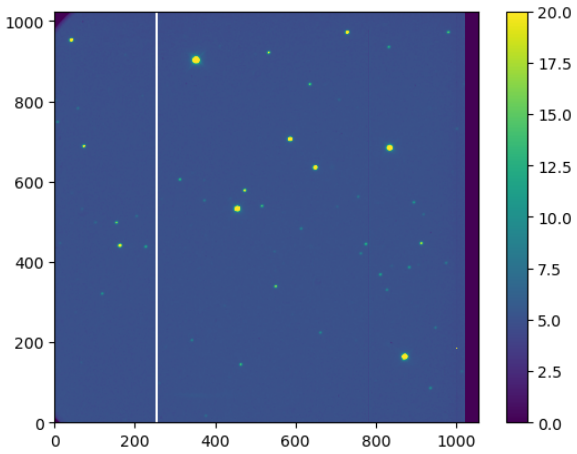


Figure 1: Calibrated Science Image with Units in Counts/Second

3 Data Reduction

3.1 Bias, Flat, and Units Corrections

To process the ccd data into a clean image with units in electron counts per second, we had to take eleven zero-second exposure bias images to correct for variations across the sensor, as well as eleven exposures of a flat image of either the twilight sky or an evenly lit screen to correct for variations in the way light is collected across the sensor. This had to be done at the beginning of every night's observing to have accurate corrections for each night. We also took five exposures of each science and calibration target so that we could later median combine them to remove any random irregularities such as cosmic rays or changing cloud cover. This was also done with the eleven bias and flat images to remove irregularities from them.

For both the combined science and calibration data, I then subtracted the counts of the combined bias, correcting any inherent variations in the sensor. This was also done to the combined flats data, achieving the same effect for the flats. I then normalized this bias-corrected flat and divided the bias-corrected science and calibration images by it, correcting for any differences in light capture across the sensor without skewing the count numbers.

To change units from counts on the ccd to electrons per second, I multiplied the raw numbers of the bias and flat corrected science and calibration images by the sensor's ratio of photons hitting the sensor and producing an electron to registered counts, which was 1.8, and then I divided each image by their respective exposure times, correcting their units to electrons per second. The figure above is an example of a fully processed science image from Oct. 6 (see Figure 1).

3.2 Stellar counts/second measurements with error bars

To get the counts per second of the target science or calibration star, I had to first make a smaller image that only contained the target star and about a star-width or two of empty space on all sides, which ended up being fifty by fifty pixels. I then made four other images of the same size in different parts of the image with no stars and median combined them to get a consistent image of the background. With this, subtracting the sum of count rates of the median-combined background image from the sum of count rates of the cropped image of the star gives the total count rate from only the star. In order to calculate the error on this count rate, the signal-to-noise ratio, I used equation 2 (see Section 4.2). Below are the cropped images of the science and calibration stars (see Figures 2, 3, 4, 5).

The table below displays the count rates of each star for each night's set of images (see Table 1).

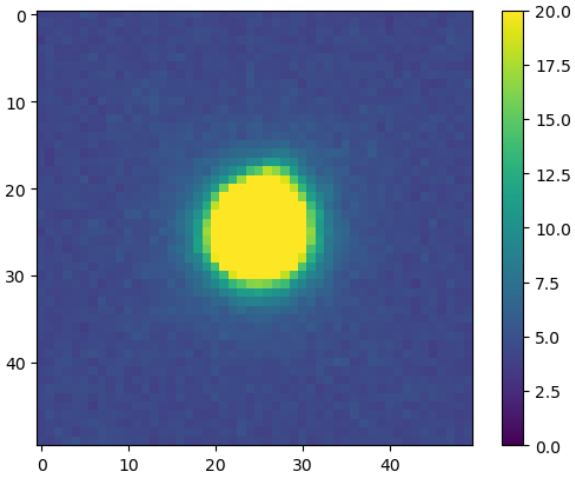


Figure 2: V0509 Peg, Oct. 4 2024

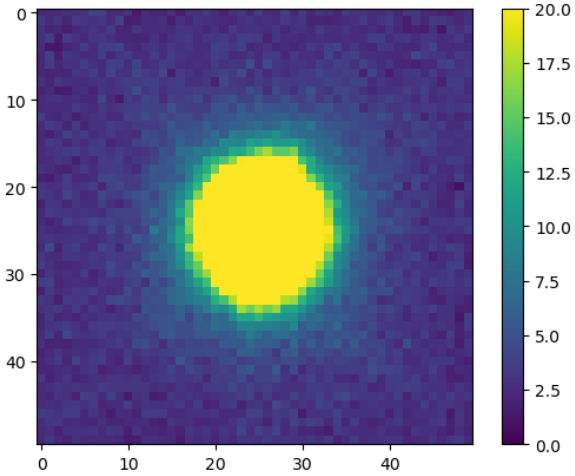


Figure 3: SA41-654, Oct. 4, 2024

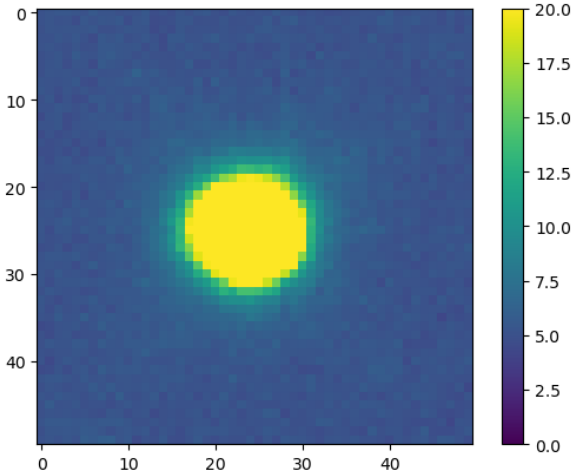


Figure 4: V0509 Peg, Oct. 6, 2024

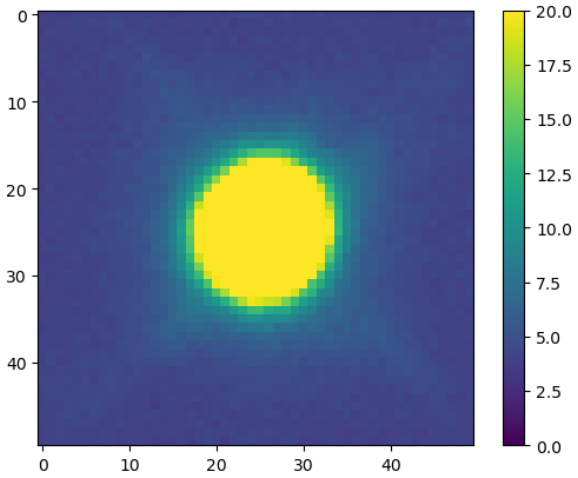


Figure 5: SA41-654, Oct. 6, 2024

Table 1: Count Rates of Each Star and Date of Observation

Star	Date of Observation	Exposure Time (s)	Count Rate (e^-/s)
V0509 Peg	Oct. 4, 2024	50	1401.862 ± 8.291
SA41-654	Oct. 4, 2024	15	40732.439 ± 8.498
V0509 Peg	Oct. 6, 2024	55	5516.328 ± 8.142
SA41-654	Oct. 6, 2024	85	39664.415 ± 8.990

4 Analysis

4.1 Estimating science star's magnitude

The calculated Vega-system magnitudes of the science star (V0509 Peg) as well as its count rates are given in the table below along with the dates the groups of exposures were taken on (see Table 2).

Having calculated the total count rates from both the science and calibration stars and the signal to noise ratio (see Section 4.2), I used the equation below (see Equation 1) as well as the known Vega-system magnitude of our calibration star (SA41-654, magnitude 10.038) to calculate the Vega-system magnitude of our science star.

$$m_{science} = m_{calib} - 2.5log_{10}(R_{science}) + 2.5log_{10}(R_{calib}) \tag{1}$$

Where $R_{science}$and R_{calib} are the calculated count rates from the science star and calibration star respectively for a given night (given in Table 1).

4.2 Error Analysis

The signal to noise ratio functions as the uncertainty in the count rate and was calculated using the equation below (see Equation 2).

$$SNR = S_y t/(S_y t + B_y t A_\Omega + RN^2_{e^-} N_{pix})^{1/2} \tag{2}$$

Where $S_y t$ is the total count rate from the cropped array with the star in it, $B_y t A_\Omega$ is the total count rate from the average background array of the same size, and $RN^2_{e^-} N_{pix}$ is the total readnoise in the

Table 2: Calculated Vega-System Magnitudes of V0509 Peg

Date Taken (mm/dd/yyyy)	Count Rate (e^-/s)	Calculated Magnitude
10/04/2024	1401.862 ± 8.291	13.696 ± 0.006
10/06/2024	5516.329 ± 8.142	12.180 ± 0.002

cropped image of the star.

The uncertainty in the magnitude ($\sigma_{m_{science}}$) was calculated using this equation (see Equation 3).

$$\sigma_{m_{science}} = ((2.5/ln(10))^2((\sigma_{R_{science}}/R_{science})^2 + (\sigma_{R_{calib}}/R_{calib})^2))^{1/2} \quad (3)$$

Where $\sigma_{R_{science}}$ is the uncertainty in the total count rate of the science star (its SNR), $R_{science}$ is the total count rate of the science star, $\sigma_{R_{calib}}$ is the uncertainty in the total count rate of the calibration star, and R_{calib} is the total count rate of the calibration star.

5 Conclusion

Overall, the results I got match up well with the expected results, with the expected magnitude being 12.86 and my experimental magnitudes coming out to 13.696 ± 0.006 on Oct. 4 and 12.180 ± 0.002. This puts the Oct. 4 magnitude 6.5% greater than the expected and the Oct. 6 magnitude only 5.3% lower than expected. These experimental magnitudes are very close to the expected magnitude, which gives me confidence that the data was recorded and processed properly. Potential sources of error could have been environmental conditions affecting the counts of specific images, needing to use less consistent flats taken on the dome screen instead of at twilight, or another reason not listed. If this same process were to be applied to several more days' observations of the same variable star, we could calculate its period of fluctuation in brightness. Since the star we chose is an RR Lyrae, we could then use the RR Lyrae period-luminosity relationship to calculate its luminosity, and with that we could calculate its distance from Earth. Its distance could then be used for many importaant purposes, including possibly calculating the rate of expansion of the universe.